

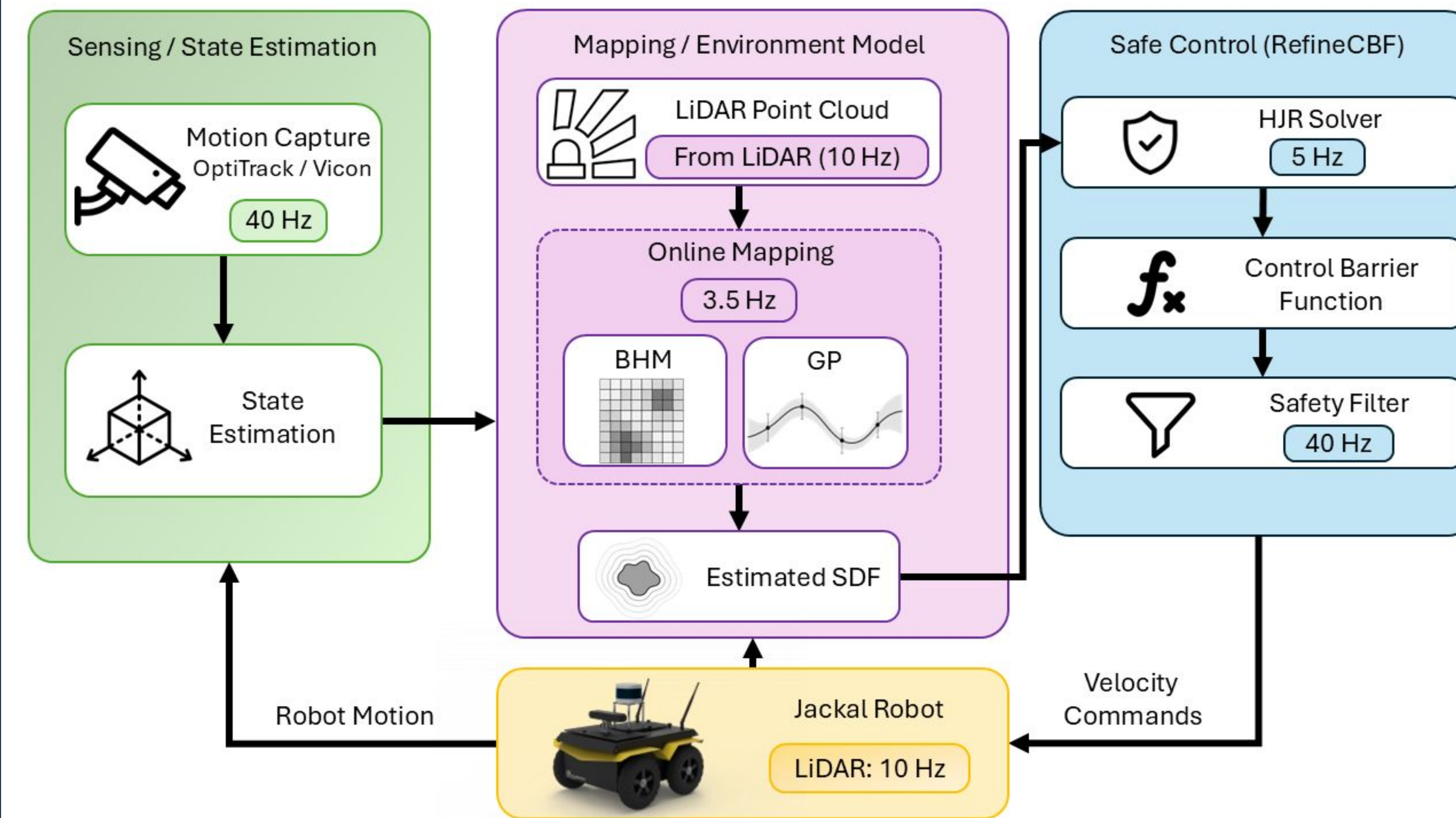
Motivation and Background

- Robots operating in unknown environments must ensure safety from noisy, partial sensor measurements.
- Hamilton–Jacobi (HJ) reachability** provides formal safety guarantees, but scales poorly with state dimension.
- Learned Control Barrier Functions (CBFs)** can provide computationally efficient safety constraints, but may not satisfy formal barrier conditions.
- We learn a **smooth CBF** from **LiDAR** using **Gaussian Process (GP) regression** and **Bayesian Hilbert Maps**, then refine it with HJ reachability to recover formal safety guarantees.

Method

- Learn:** GP regression and Bayesian Hilbert Maps (BHM) produce a **smooth Signed Distance Function (SDF)** with probabilistic **uncertainty bounds**.
- Refine:** Hamilton-Jacobi Reachability transforms the learned SDF into a **valid Control Barrier Value Function**
- Update:** New LiDAR measurements **continuously update** the model and **expand** the certified **safe region**

GP-CBF System Overview



On-board **LiDAR measurements** are used to train the **BHM** and **GP** to produce a **smooth SDF** of the environment. The SDF is then refined via HJ Reachability into a **valid CBF** and enforced online via a safety filter. New LiDAR measurements **update** the **SDF** as the robot navigates.

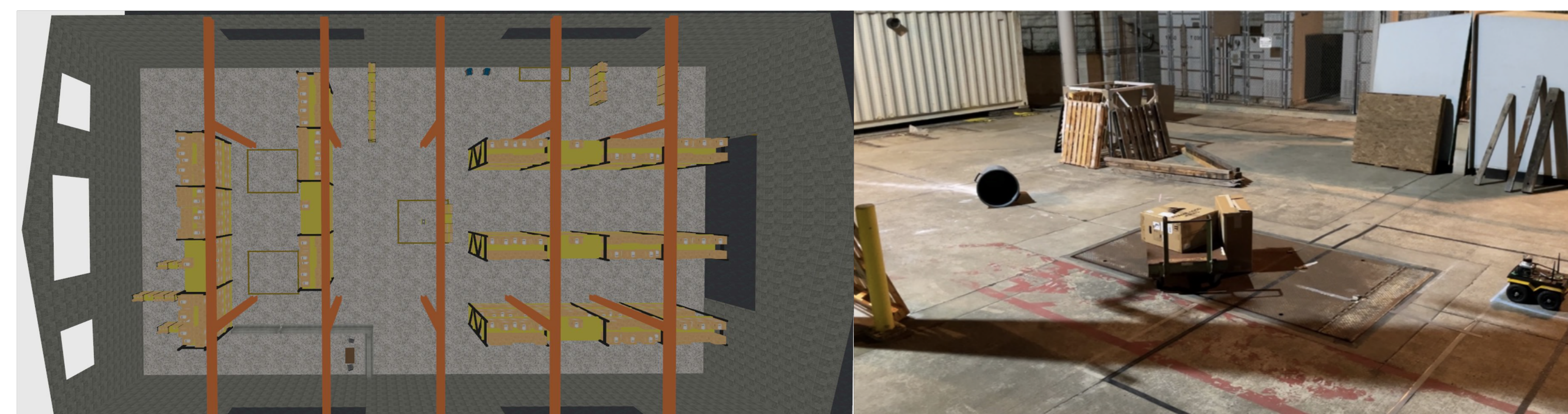
Evaluation

Experiments were conducted in **hardware** and **simulation**, consisting of a navigation task in **a priori unknown** environments.

5 baseline methods were used to evaluate the effect of each component of the system: SDF generation, uncertainty estimation via a GP, CBF refinement, and use of a path planner.

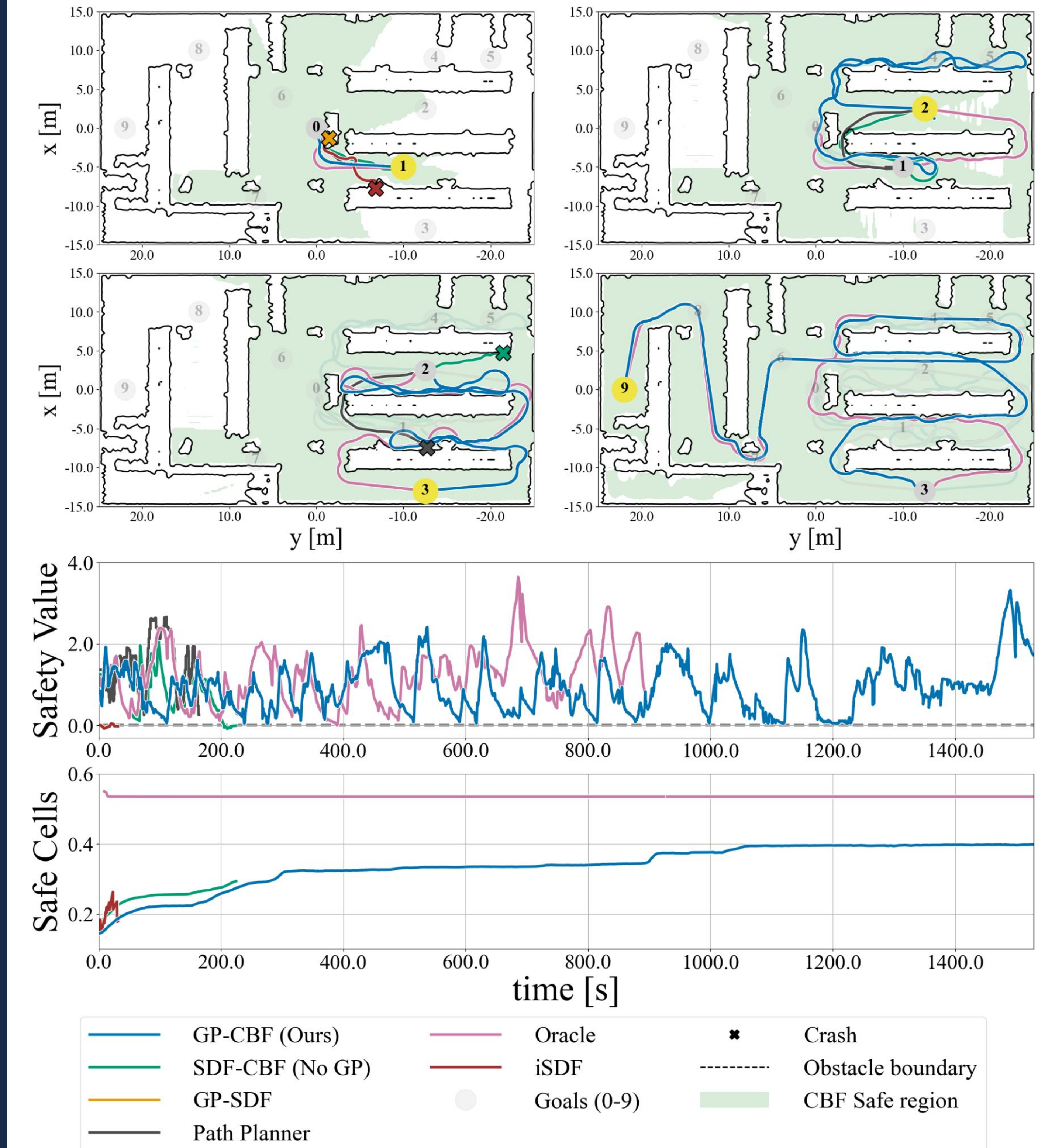
Simulation Environment

Hardware Environment



Method	SDF	GP	CBF	Planner
Oracle	✓	✓	✓	
GP-SDF	✓	✓		
SDF-CBF (No GP)	✓		✓	
D* Lite		✓		✓
iSDF	✓		✓	
GP-CBF	✓	✓	✓	

Results



Performance of each control policy in simulation, depicting four snapshots of the learned safe set and comparison trajectories throughout the navigation task.

Control Policy	Goals Reached	Elapsed Time (s)	Distance Traveled (m)	Min Safety Value
Oracle	9/9	896	262	0.25
GP-SDF	0/9	7	2	0.00
SDF-CBF	2/9	178	55	0.00
D* Lite	2/9	98	57	0.00
iSDF	0/9	43	12.5	0.00
GP-CBF (Ours)	9/9	1780	422	0.05

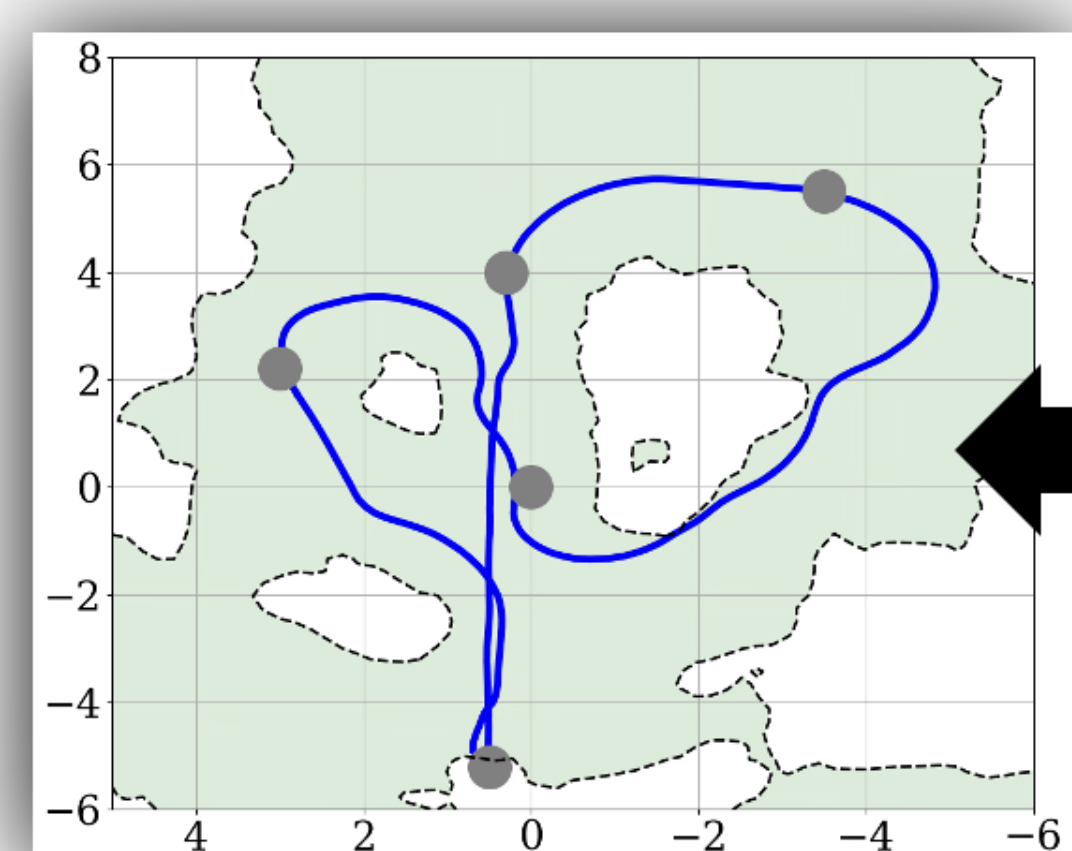
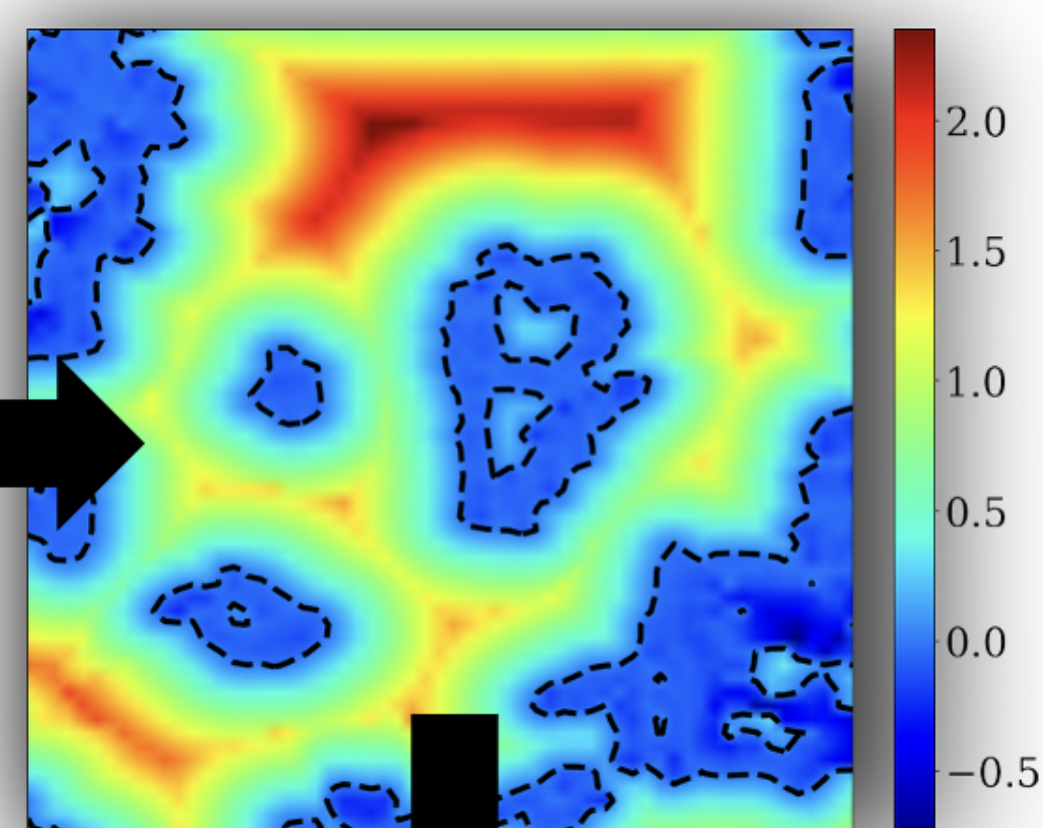
Major Takeaways:

- GP-CBF reached all goals while maintaining positive safety value.**
- Baseline methods frequently violated safety or failed to complete the task.**
- The certified safe region expands online as new LiDAR measurements arrive**

Environment



SDF



Safe Control

CBVF

